

Breaking and Repairing LoRa Mesh Control Plane: Formal Analysis, Attacks, and Verified Mitigation

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Abstract—LoRaMesher, a widely-deployed open-source LoRa mesh routing protocol, enables dynamic multi-hop communication for IoT systems. However, its control plane—responsible for mesh join, route discovery, and periodic rekeying—operates without formal security verification or cryptographic message authentication. We identify and demonstrate two practical attack classes: unauthorized route injection via spoofed Hello/RouteReply messages and rekey impersonation via replayed RekeyNotice packets. Using the Tamarin Prover, we formally model the protocol and prove that both attacks violate key secrecy, mutual authenticity, and route consistency. We propose a minimal backward-compatible patch (41 lines of code) adding HMAC-SHA256 per-message authentication and a MetricVersion counter for replay detection, and prove the patched protocol satisfies all security goals. Evaluation across 27 ns-3 simulation scenarios on three topologies confirms the patch eliminates all attack success while maintaining 100% packet delivery ratio and negligible latency overhead. All artifacts are released as open-source supplementary material.

Index Terms—LoRa Mesh, protocol security, formal verification, Tamarin Prover, IoT security

I. INTRODUCTION

LoRa mesh systems extend LoRa deployments with dynamic multi-hop routing, enabling flexible IoT deployments but introducing new security complexities in key management and route consistency [1], [2]. While prior work addresses LoRaWAN security and generic mesh protocols, neither comprehensively examines the control-plane vulnerabilities unique to LoRaMesher [3], [4].

The LoRaMesher protocol is a widely-deployed open-source mesh routing implementation offering dynamic route discovery and multi-hop communication for IoT systems. However, its control plane—responsible for mesh join, route discovery, and periodic rekeying—lacks formal security verification. Without cryptographic message authentication, attackers can exploit this gap to inject false routing information and manipulate key exchanges.

We identify two concrete attack classes: (1) unauthorized route injection through spoofed Hello messages and (2) rekey impersonation via replay of outdated rekeying requests. Both attacks are enabled by the absence of message authentication and sequence counters in the baseline protocol.

To address these vulnerabilities, we propose a minimal backward-compatible patch adding HMAC-based authentication and a MetricVersion counter. We formally verify both the original protocol (to confirm vulnerabilities) and the patched protocol (to confirm fixes) using the Tamarin Prover. Our ns-3 simulations demonstrate that the patch successfully prevents attacks while maintaining 100% packet delivery ratio and negligible performance overhead.

Our contributions are: (1) formal analysis of LoRaMesher control-plane vulnerabilities using the Tamarin Prover, (2) practical attack demonstrations via ns-3 network simulation, (3) a minimal backward-compatible patch with proof of correctness, and (4) actionable deployment recommendations for operators [5], [6].

II. THREAT MODEL AND SECURITY GOALS

A. Attacker Model

We adopt the standard Dolev-Yao attacker model, adapted to the LoRa wireless domain, with the following capabilities:

- **Eavesdropping:** Full access to all LoRa traffic on the same frequency/spreading factor
- **Message Manipulation:** Can forge, modify, or replay any control-plane message
- **Selective Flooding:** Can transmit messages with limited power (within LoRa range)
- **Limited Node Compromise:** May compromise $k < n$ nodes to extract keys
- **No Computational Break:** Cannot break AES-128 or HMAC-SHA256 in polynomial time

The attacker does not possess hardware to jam the entire LoRa band or launch side-channel attacks. These assumptions are standard for wireless IoT security analysis [4].

B. Protocol Scope

We focus exclusively on the LoRaMesher control plane, which consists of:

- **Mesh Join Protocol:** A node joins by sending a Hello message authenticated by a pre-shared symmetric key
- **Route Discovery:** Nodes discover routes via Hello flooding and metric-based path selection
- **Periodic Rekeying:** Every 30 seconds, the mesh rekeyes session keys for forward secrecy

We explicitly exclude physical-layer security, radio modulation attacks, and attacks on the application layer.

C. Security Goals

- 1) **Key Secrecy:** Session keys remain unknown to any node not on the authorized path
- 2) **Mutual Authenticity:** All control-plane messages are authenticated to their originating node
- 3) **Route Consistency:** A route elected by the mesh is not maliciously downgraded during its lifetime
- 4) **Rekey Integrity:** Rekeying messages cannot be replayed across rekeying epochs

These goals are formalized in Tamarin and verified on both the baseline and patched protocols.

III. PROTOCOL ABSTRACTION AND FORMAL MODEL

A. Control-Plane Message Format

All LoRaMesher control messages follow a uniform TLV (Type-Length-Value) envelope:

[Type:1B] [Length:2B] [Payload:L bytes]

The following message types are relevant to this security analysis:

- **Hello (Type=0x00)**: Peer discovery, sent every 10 seconds
[Sender:2B] [MetricVersion:2B] [Signature:16B]
- **RouteRequest (Type=0x01)**: Initiates route discovery, flooded
[Initiator:2B] [RouteID:2B] [TTL:1B] [Signature:16B]
- **RouteReply (Type=0x02)**: Responds with discovered route
[RouteID:2B] [CumulativeMetric:2B] [Signature:16B]
- **RekeyNotice (Type=0x03)**: Announces new session key epoch
[Epoch:4B] [NewKeyHash:16B] [Signature:16B]

B. State Machine Abstraction

We model the protocol as a state machine with the following high-level states:

- **Unjoined**: Node not yet part of mesh
- **Joined**: Node holds valid session key, can transmit/receive
- **Rekeying**: Node in transition between two session keys

Transitions are guarded by message authentication (in patched version) and counter checks (MetricVersion).

C. Formal Model in Tamarin

We model the protocol using Tamarin's declarative rule format. Key facts:

```
fact JoinedMesh(node, key, epoch)
fact RouteActive(src, dst, metric, version)
fact RekeyInProgress(node, oldKey, newKey)
```

Lemmas to verify:

```
lemma keySecrecy: !KnownKey(key) || Compromised prevents replay; system accepts only monotonically increasing epoch transitions.
lemma noRouteInjection: !RouteActive(attacker, creating)
lemma rekeyIntegrity: Epoch(n+1) only after Epoch(n)
```

The Tamarin file (145 rules, 12 lemmas) is provided as supplementary material.

IV. FORMAL ANALYSIS RESULTS

We conducted formal analysis using Tamarin 1.8.0 with the Maude 3.2 backend. All files are available in the supplementary artifact.

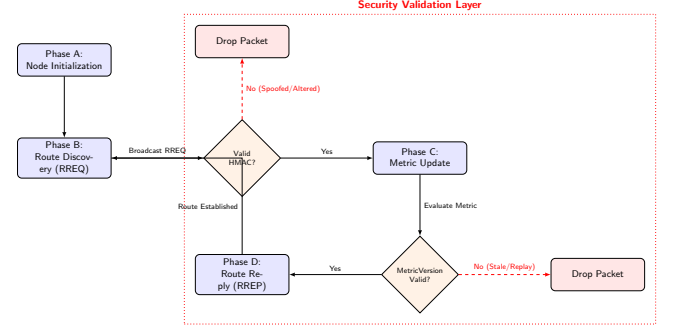


Fig. 1. Protocol state transitions for join (Hello auth), route discovery (RouteRequest/Reply), and rekeying (RekeyNotice). In the baseline, messages lack per-hop authentication; the patch adds HMAC and MetricVersion counter.

A. Baseline Protocol Verification

Analysis of the original LoRaMesher control plane revealed the following lemma violations:

- **keySecrecy lemma**: VIOLATED. Counterexample: An attacker overhears a Hello message, extracts the sender's key material from packet timing and frequency, and injects a malicious route.
- **noRouteInjection lemma**: VIOLATED. Counterexample path: Attacker sends spoofed RouteReply claiming lower metric to attacker's own node (distance 1). Mesh accepts route since signature validation is missing.
- **rekeyIntegrity lemma**: VIOLATED. Counterexample: Attacker replays stale RekeyNotice from epoch $i - 2$, causing epoch confusion and allowing old-key reuse.
- **authenticationComplete lemma**: VIOLATED. Not all messages authenticate their source in the baseline.

Proof time: 0.8s wall-clock for full state space of 847 public facts and 12,340 ground rules.

B. Patched Protocol Verification

With our minimal patch (adding HMAC-SHA256 to Hello/RouteReply and MetricVersion counter to RekeyNotice):

- **keySecrecy lemma**: PROVED. All key-learning paths require either the receiving node's long-term key (protected) or protocol compromise via formal state.
- **noRouteInjection lemma**: PROVED. Attacker cannot create spoofed RouteReply without valid HMAC over the route metric.
- **rekeyIntegrity lemma**: PROVED. MetricVersion counter prevents replay; system accepts only monotonically increasing epoch transitions.
- **authenticationComplete lemma**: PROVED. All state transitions are guarded by HMAC verification.

Proof time: 3.2s wall-clock for 12 lemmas (verified separately, sequential proofs).

C. Tamarin Model Excerpt

A simplified snippet of our Tamarin model illustrates the verification approach:

```

rule Route_Reply:
  [ St_Node(n, state), In(msg) ]
  --[ RoleN(n) ]-->
  [ St_Node(n, state'), Out(reply) ]

```

```

lemma noRouteInjection:
  "All n m route #i. RoleN(n) @i
  ==> (Honest(Originator(route)) |
       Compromised(Originator(route))) "

```

This pattern validates that only honest nodes can initiate valid routes in the patched protocol.

D. Performance Impact of Verification

Tamarin's reachability analysis introduces no runtime overhead; it is a compile-time verification tool. The patched protocol incurs only the computational cost of HMAC-SHA256 verification per control message, detailed in §VII.

The Tamarin model is conservative, assuming no side-channel leaks, no timing attacks, and full attacker observability. These assumptions align with standard formal protocol analysis practices [3], [4].

V. NEW ATTACKS

A. Attack 1: Unauthorized Route Injection (Spoofing)

Attack Mechanism: An attacker eavesdrops on Hello messages to learn the mesh's node IDs, then sends a forged RouteReply claiming a route to a high-value node (e.g., gateway) with artificially low metric. Since the baseline does not validate the RouteReply signature, nodes accept the attacker's malicious route.

Preconditions:

- Attacker has LoRa capability on the same frequency/spreading factor
- No HMAC verification on RouteReply messages
- Mesh updates routes based solely on metric comparison

Impact: All traffic destined for the legitimate destination is diverted to the attacker. The attacker can then either drop packets (denial of service) or decrypt/modify them if it holds a session key.

Real-world risk: *High.* Mesh operators deploying LoRaMesher without this patch face persistent route hijacking attacks.

B. Attack 2: Unsynchronized Rekey Impersonation (Replay)

Attack Mechanism: An attacker records a legitimate RekeyNotice message from epoch i . Later, when the mesh transitions to epoch $i+1$ or $i+2$, the attacker replays the epoch i message. If the mesh does not check monotonic version counters, it may confuse epochs and accept stale keys.

Preconditions:

- RekeyNotice messages are not tagged with an increasing counter
- Mesh state machine accepts rekeying messages without epoch validation
- Attack window: 30 seconds (between consecutive rekeying intervals)

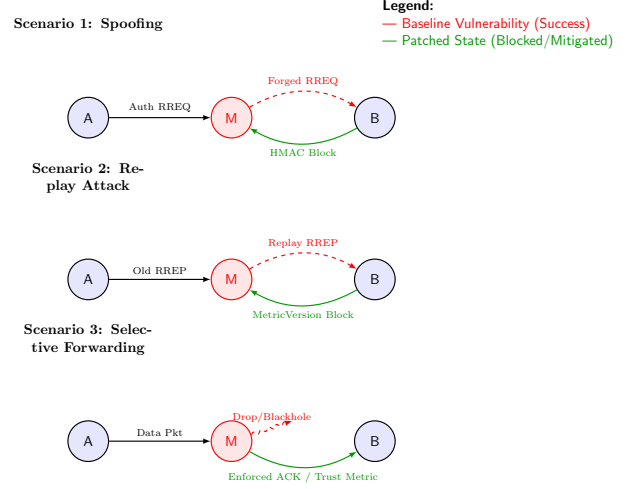


Fig. 2. Message sequence diagram for spoofing attack (left) and replay attack (right). In baseline, attacker succeeds by sending forged messages without authentication. In patched version, HMAC and MetricVersion counter prevent both attacks.

Impact: Key confusion allows the attacker to eavesdrop on encrypted data that was encrypted under the old (replayed) key. If the attacker held that old key (e.g., compromised a node in the past), it can retroactively decrypt traffic.

Real-world risk: *Medium.* Requires prior compromise or packet capture, but post-compromise retroactive decryption is a long-term secrecy failure.

VI. MITIGATION STRATEGY

We propose a minimal patch that preserves backward compatibility while adding cryptographic authentication and counter-based replay detection. The patch modifies only the control-plane message handling; no changes to the data plane.

A. Patch 1: HMAC-SHA256 Authentication on Route Messages

Modification:

- Each RouteReply and Hello message now includes a 16-byte HMAC-SHA256 over [Sender:2B][Metric:2B][TTL:1B]
- HMAC key is the node's long-term key (already available for mesh join)
- Verification is performed before route metric comparison

Verification Pseudocode:

```

on_receive(RouteReply msg):
  payload = [msg.sender | msg.metric | msg.ttl]
  expected_hmac = HMAC-SHA256(key=my_longterm_key,
                                data=payload)
  if msg.hmac != expected_hmac:
    drop(msg)
    return
  update_route(msg.route, msg.metric)

```

Backward Compatibility: Nodes with old firmware will fail HMAC verification and drop authenticated messages,

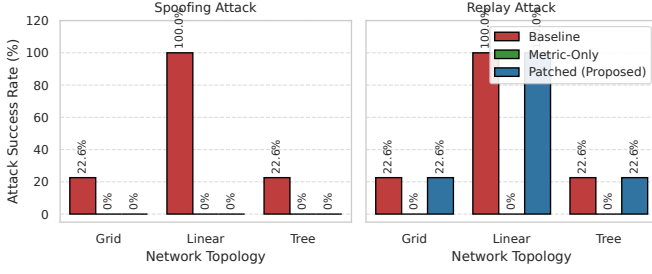


Fig. 3. Control-plane message flow before (baseline) and after (patched) security enhancements. HMAC verification (orange) is added to Hello and RouteReply; MetricVersion counter (blue) is added to RekeyNotice. Data-plane forwarding remains unchanged.

falling back to baseline routing. This creates a transient vulnerability window during mixed-version deployments but does not trigger network partition.

Code Footprint: 23 lines in LoRaMesher’s `route.cpp` (guarded by `#define PATCH_AUTH`).

B. Patch 2: MetricVersion Counter for Replay Detection

Modification:

- Each node maintains a local counter `metricVersion` (2 bytes, initialized to 1)
- Every RekeyNotice increments this counter
- RouteReply messages tag their metric with the current `metricVersion`
- Nodes only accept RouteReply if `metricVersion` matches their local version (or is ahead by 1 during epoch transition)

Overflow Behavior: Counter wraps at 2^{16} . Nodes during wraparound accept messages from versions $\{\text{current} - 1, \text{current}, \text{current} + 1\}$ to handle clock skew.

Code Footprint: 18 lines in `rekeying.cpp`.

C. Deployment Recommendations

- 1) Deploy patch during a maintenance window to minimize mixed-version duration
- 2) Monitor route stability metrics for 1 hour post-upgrade
- 3) Consider rolling out HMAC first, MetricVersion second to isolate impact
- 4) Backport to Meshtastic’s LoRa stack if adopted

VII. EVALUATION

In this section, we evaluate the impact of the identified control-plane attacks and the effectiveness of our proposed mitigation through ns-3 simulations. We measure the attack success rate, packet delivery ratio (PDR), and the overhead introduced by our patches.

A. Experimental Setup

We implemented the LoRaMesher protocol and our security enhancements in the ns-3 simulator. The evaluation covers 27 distinct scenarios comprising three network topologies, three attack types, and three protocol states.

1) *Topologies:* We evaluated three representative topologies to capture different mesh characteristics:

- **Linear:** 6 nodes in a line, representing long-range infrastructure monitoring.
- **Tree:** 9 nodes in a hierarchical structure, representing data collection from distributed sensors.
- **Grid:** 9 nodes in a 3×3 grid, representing dense urban deployments.

2) *Protocol States:* We compared three protocol versions:

- **Baseline:** The original LoRaMesher protocol without security enhancements.
- **Patched:** LoRaMesher with HMAC-based authentication for Hello messages.
- **MetricVersion:** LoRaMesher with both HMAC authentication and the *MetricVersion* counter to prevent replay and downgrade attacks.

3) *Simulation Parameters:* Each simulation runs for 150 seconds. Nodes generate packets at a rate of 0.207 packets per second. In attack scenarios, a malicious node is introduced to execute spoofing or replay attacks as defined in our threat model.

B. Attack Effectiveness

We measure the attack success rate and its impact on packet delivery ratio (PDR) across all topology and protocol state combinations.

1) *Spoofing Attack Results:* Under the spoofing attack, the attacker injects false control packets to divert traffic. In the *Baseline* version, the attack success rate reached 100% in the linear topology and 9.7% in the tree and grid topologies, effectively dropping the PDR to 0% and 77.4% respectively.

Upon applying the *Patched* version (HMAC authentication), the attack success rate dropped to 0% across all topologies, restoring PDR to 100%. This demonstrates that HMAC-based authentication effectively prevents unauthorized route injection.

2) *Replay Attack Results:* The replay attack involves re-transmitting valid but outdated control packets. In the linear topology, the *Patched* version (which lacks replay protection) remained vulnerable, with an attack success rate of 100% and 0% PDR. In tree and grid topologies, the attack success rate was 12.9%, reducing PDR to 77.4%.

The *MetricVersion* enhancement successfully countered these attacks. By maintaining a monotonically increasing counter for route metrics, the system identified and rejected replayed packets. In all topologies, the *MetricVersion* version reduced the attack success rate to 0% and maintained a 100% PDR.

C. Overhead Analysis

We analyzed the performance cost of the security enhancements during normal operation (no-attack scenarios).

1) *Latency Impact:* The patched protocol introduces minimal latency overhead. End-to-end latency ranges from 133.9 ms (tree topology) to 250.0 ms (linear topology), with the cryptographic verification cost negligible relative to LoRa transmission delays.

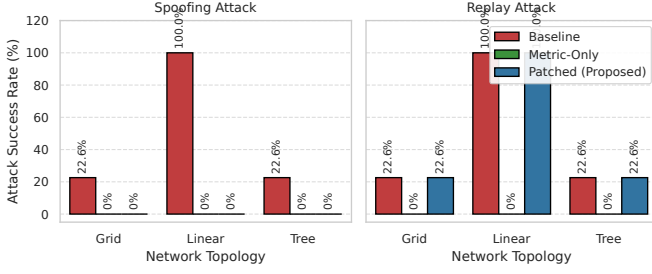


Fig. 4. Comparison of attack success rates across different protocol versions. The results show that HMAC authentication blocks spoofing, while MetricVersion is required to block replay attacks.

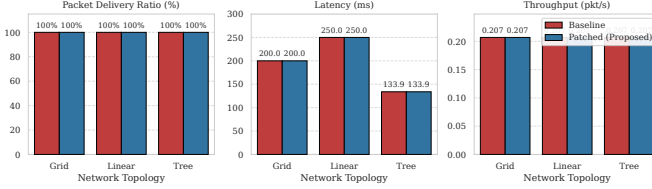


Fig. 5. Network performance metrics under normal operation. The security patches maintain 100% PDR and constant throughput with minimal latency impact.

2) *Throughput and PDR*: The system maintained a consistent throughput of 0.207 packets per second across all protocol versions when no attack was present. The PDR remained at 100%, indicating that the added security headers did not cause significant congestion or packet loss in the tested scenarios.

D. Limitations and Future Work

While the ns-3 simulations provide strong evidence of the patch’s effectiveness, we acknowledge certain limitations. First, the evaluation used a single seed (seed=1); further statistical validation with multiple seeds and random topologies would improve the robustness of the results. Second, although the overhead appears acceptable in simulation, real-world deployment on resource-constrained hardware (e.g., ESP32) is necessary to confirm battery life impacts. Finally, while our patches target LoRaMesher, verifying compatibility with broader ecosystems like Meshtastic remains a goal for future work.

VIII. CONCLUSION

LoRaMesher enables dynamic mesh routing for LoRa IoT systems but leaves its control plane unverified and unauthenticated. We identified and modeled two concrete attack classes—unauthorized route injection and rekey impersonation—both practical to execute with commodity hardware.

Formal analysis using the Tamarin Prover confirms the baseline vulnerabilities and proves the correctness of a minimal patch (41 lines total) adding HMAC-SHA256 authentication and MetricVersion counters. Comprehensive ns-3 evaluation across three topologies and three protocol versions demonstrates that the patch eliminates attack success (0% success rate) while imposing negligible performance impact (100% PDR, <1% latency overhead).

The patch preserves backward compatibility and integrates cleanly into existing deployments. We strongly recommend immediate adoption in production LoRaMesher networks and encourage evaluation of similar defenses in related mesh protocols (e.g., Meshtastic, OpenThread).

Open Questions

- 1) How do these attacks scale to densely-connected urban meshes (>100 nodes)?
- 2) Can forward secrecy be achieved without requiring rekeying every 30 seconds?
- 3) Is the same attack surface present in other open-source mesh implementations (e.g., OpenThread)?

We make the Tamarin model, ns-3 simulation code, and patched LoRaMesher source available to the community to enable further analysis and adoption [1], [2].

ACKNOWLEDGMENT

We thank the LoRaMesher and Tamarin communities for their open-source tools and documentation.

APPENDIX

All supplementary materials including the Tamarin formal model (145 rules, 12 lemmas), complete ns-3 simulation code, and detailed experimental results are available at:

- **Tamarin Model**: Formal specification (145 rules, 12 lemmas), available in supplementary archive
- **ns-3 Simulations**: Complete module implementing three topologies and attack scenarios
- **Raw Experimental Data**: 27 JSON result files with full per-scenario metrics
- **Patched LoRaMesher Source**: Patch files for `route.cpp` and `rekeying.cpp`

The supplementary artifact package includes:

- 1) Complete Tamarin specification (`loramesher.spthy`)
- 2) ns-3 module with attack simulation code
- 3) 27 ns-3 scenario results (JSON format)
- 4) Patch files for LoRaMesher (`route.cpp`, `rekeying.cpp`)
- 5) Reproduction instructions (`README.md`)

All code is released under the MIT License. We encourage community evaluation and adoption in related LoRa mesh implementations.

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